FINAL HANDOVER · VERSION 1.3.6 (VERSIONCODE 17)

SummaRace v3 Project Documentation

Compiled 2026-11-06 · Android app `com.summarace.app` · source github.com/orbitsdev/summaracev3

SummaRace is an offline reading game for Grade 4 learners: read a story, race to collect its parts, put them in order, and sum it up in 2–3 sentences. This file puts every project document in one place.

Contents

1. Product Requirements (PRD)
2. Technical Requirements (TRD)
3. App Flow
4. Design Brief (system design)
5. Data / Backend Schema
6. Implementation Plan
7. Assets, Sounds and Credits
8. Handover and known open items

PART 1 · PRODUCT REQUIREMENTS DOCUMENT

Product Requirements Document

SummaRace v3 · version 1.3.6 · Android app `com.summarace.app`

1 Overview

This document says what SummaRace does, feature by feature, and how each feature is expected to behave. It describes the game as it is built today, not a wish list.

SummaRace is a reading game for Grade 4 learners. For each story the learner **reads** it page by page, **races** through its story parts in a 3D runner, puts the parts **in order**, and then **sums it up** in 2–3 sentences. The tagline on the title screen is "Read it · Race it · Sum it up!". Teacher Mira, a cartoon teacher, guides the learner with spoken lines the whole way.

2 Product goal

Help Grade 4 learners practise reading comprehension and summarising with the **Somebody – Wanted – But – So – Then** (SWBST) method, in a game they want to keep playing, and give the teacher a record of every finished level that they can open in a spreadsheet.

The game has to work on low-cost Android phones in a classroom with no internet and no accounts.

3 Users

Learner

A Grade 4 pupil. Types their name and section, plays the stories in order, earns stars and coins.

Teacher

Hands out the phones, reads the records screen, exports the CSV, can erase or bring back records.

Research team (client)

Wrote the 30 stories, questions and SWBST answers. Uses the CSV results for the study.

4 Content scope

- **30 stories** in **10 sessions**. Each session has one easy, one average and one hard story (levels 1–30).
- Each story has **5 pages**, **5 page questions** (one per page), **1 main idea question**, and 5 **SWBST story parts** with right and wrong answers.
- The 30 levels sit in **5 worlds** of 6 levels each: Sunny Meadow, River Bend, Story Town, Palm Beach and Summit Peak.
- The pages, page questions, SWBST answers and main idea come word for word from the research team. The wrong answer choices, race wording and Teacher Mira's example summaries were written for the game.

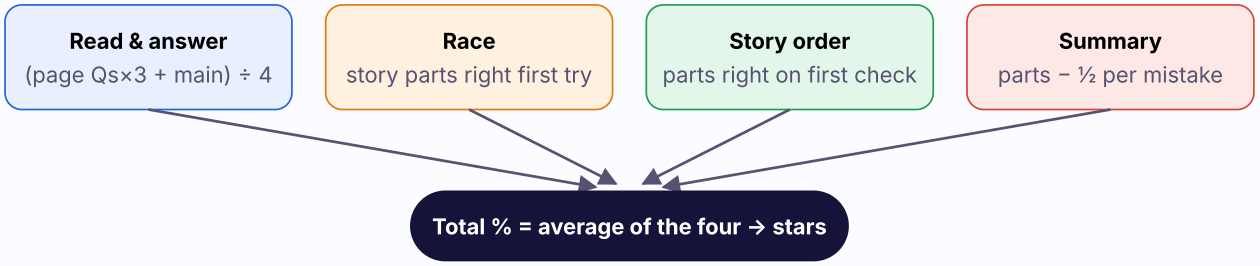
5 Features

Feature	Description	Expected behaviour
Title screen	Animated title with Teacher Mira, a PLAY button and Credits.	PLAY opens "Who is playing?". Credits opens the credits popup.
Learner profiles	Name (required, max 40 characters) and section (optional, max 30).	A new name and section makes a new profile on this device (matching ignores capital letters). Returning learners tap their name under "Or continue as". Progress is saved per profile.
World map	A scrolling path of 30 level nodes with a session bar (sessions 1–10) and a "go to my level" button.	Levels unlock one by one. Tapping a locked level shows a card that sends the learner to their open level. Stars and best score show on finished levels.
Read	One page at a time with a world picture, page dots and a Listen button.	Mira says a short tip and then reads the page aloud with a recorded voice. "Answer the question" goes to that page's question.
Page questions	3 answer choices in a new random order every time.	A wrong answer is never revealed. After a wrong answer Mira offers "Read the story" or "Try again". After 2 wrong tries the story opens with the right part highlighted. Only a right first try counts for the score.
Main idea	After the last page, 4 choices: which one tells what the whole story is about.	Same retry rules as page questions. Scored right or wrong on the first try.
Race	A 3D endless runner: the learner's runner (Max or Mia) escapes a police officer and dog.	Each SWBST part becomes a question at the top with answers A, B and C shown in the 3 lanes. The learner swipes into the lane of the right answer and runs through its gate. Right = boost and the patrol falls back. Wrong = the patrol comes closer and the question comes back later. Trains and barriers must be dodged (swipe left/right, up to jump, down to roll). If the patrol catches the runner, the race restarts.
Story order	Tap the 5 story parts into the Somebody, Wanted, But, So, Then slots.	Right parts turn green and stay. After the first wrong check a 30-second clock starts (it pauses while the story or Settings is open). When time runs out Mira shows the right order and the level goes on. Score = parts right on the first check.
Summary	Write 2–3 sentences in English, in your own words.	An offline checker gives feedback lines (see TRD). It passes when the summary names the main character, has most story parts and the main idea, and has no clear mistakes. After 3 tries Mira shows an example. After 4 tries the learner can choose "Finish, my teacher will check".
Results	Stars (1–3), a title, the score for each stage and coins earned.	Total ≥ 85% = 3 stars "AMAZING!", ≥ 65% = 2 stars "GREAT JOB!", otherwise 1 star "GOOD TRY!". Buttons: Map, Replay, Next. The next level unlocks.
Teacher records	A table of every finished level on this device.	Copy CSV, Download CSV (saved to `Download/SummaRace/` on Android), Share CSV (Android share sheet). "Erase all records" needs the word ERASE typed in; the last erase can be undone with "Bring back erased records".
Settings	Teacher voice, sound effects and music switches; Change player; Teacher records; Quit this story; Exit game.	Switches are saved on the device. Quit is offered during a level, Exit game only inside the Android app.
Teacher Mira	A guide docked at the bottom of every screen except the race and records.	Speaks recorded lines and shows a speech bubble. The bubble folds away after 7 seconds, or after 2 seconds if it covers a button the learner needs. Tap her to hear the line again.

Android Back button	Handled by the game, not the system.	Closes a popup first. In the race, the first Back pauses. During a level it opens the Quit popup instead of dropping progress. On the map it goes back to the title screen.

6 Scoring

The level total is the average of four stage scores:



Questions, race and story order count the first try only, so retries help the learner finish but do not raise the score. The summary counts the try that passed, minus half a part for each wrong detail. "Finish, my teacher will check" scores the summary as 0 and leaves it for the teacher.

Coins collected in the race are added to the learner's coin total. A replay keeps the best stars and best score for that level.

7 Constraints

- **Fully offline.** No server, no login, no analytics, no ads. Everything ships inside the APK.
- Android 7.0 or newer (minSdk 24), portrait only, phones with about 2 GB of RAM.
- Records live only on the device they were made on. The teacher must collect the CSV from each phone.
- The summary checker must run offline, so it uses word lists and rules, not an AI model.

8 Out of scope

- Syncing records between devices or to the cloud.
- Teacher accounts, passwords or a class dashboard.
- Languages other than English (the summary must be written in English).
- iOS builds (the web game would run in a browser, but no iOS app is made).

9 Open items

Item	Status
Two stories are both titled "In Grandfather's Day" (levels 15 and 18).	Needs the client
Profiles are matched by name and section, so two learners with the same name in the same section on one phone share a profile.	Open
Some newer Mira lines have no recorded clip and fall back to the phone's voice. The clip build script was lost.	Open

PR #4: race music plays faster as story parts are collected.	Waiting for a decision
PR #5: Story Stop voice lines. No longer needed since the race was changed to lane answers.	Can be closed

PART 2 · TECHNICAL REQUIREMENTS DOCUMENT

Technical Requirements Document

SummaRace v3 · version 1.3.6 (versionCode 17)

1 Overview

This document lists the tools, technologies and structure used to build SummaRace v3, so anyone who picks up the code knows how it fits together and what it must keep doing.

SummaRace v3 is a web game (HTML, CSS and plain JavaScript, with three.js for the 3D race) that runs inside a small Android app. The Android app is a full-screen WebView plus a JavaScript bridge for things a web page can't do on its own: the phone's voice, the clipboard, sharing, saving files and closing the app.

2 Technical goals

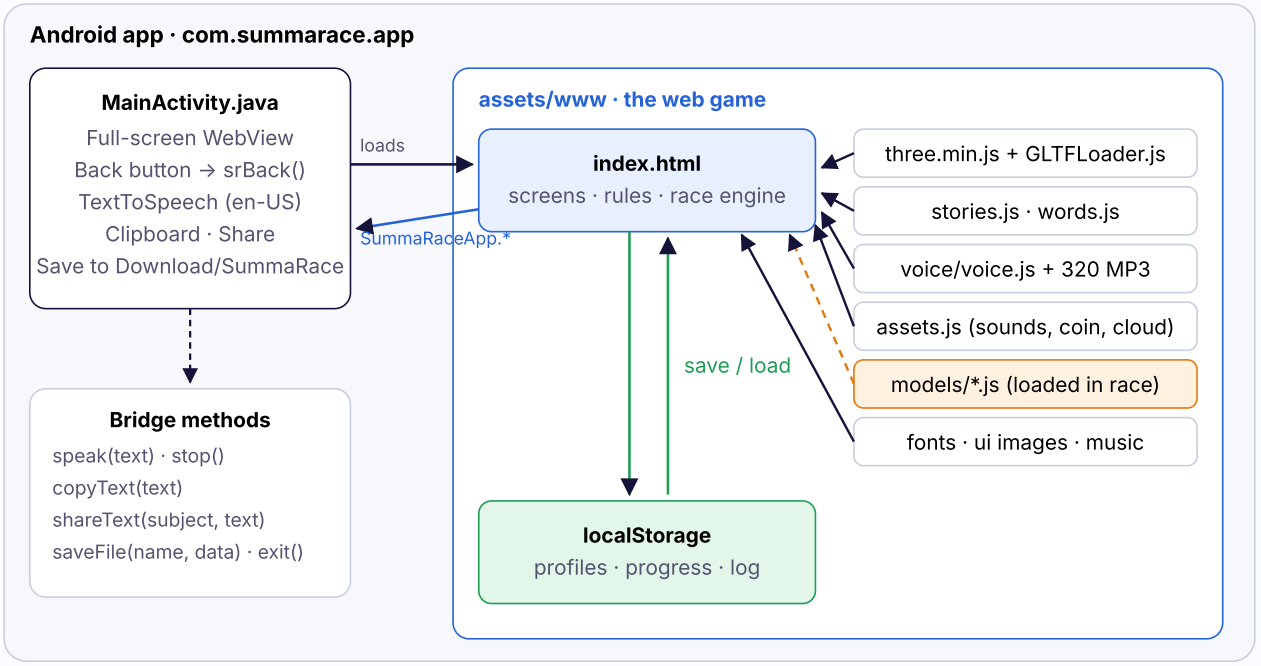
- **Works with no internet.** Every file is inside the APK. Nothing is fetched from the network.
- **Runs on low-cost phones.** Android 7.0+ (API 24), about 2 GB RAM. The 3D models only load when the race starts, and GPU memory is freed after each obstacle.
- **Easy to change.** No build step for the game. Stories are one data file. Opening the folder with a local web server runs the game in a desktop browser.
- **Safe for learners.** No accounts, no tracking, no ads, no outside links during play.

3 Technology stack

Area	Tool / technology	Purpose
Game UI	HTML5, CSS, plain JavaScript (ES2018), one `index.html`	Every screen, popup, Teacher Mira, quiz, order and summary logic. No framework.
3D race	three.js r128 (`three.min.js`) + `GLTFLoader.js`	The endless-runner race: toon shading, animated characters, trains, scenery.
3D models	GLB files stored as base64 text in `models/*.js`	Loads from inside the app with no file-access problems. Compressed with glTF-Transform 4.1.1.
Sound effects	Web Audio API, clips in `assets.js` as data URLs	Click, coin, jump, bump, good, bad, win and so on.
Music	HTML `Audio` elements, looping MP3	`menu.mp3` on all screens, `race.mp3` in the race. Ducks under Mira's voice.
Voice	320 pre-recorded MP3 clips (Kokoro TTS) + `voice/voice.js` index	Mira's lines and all 150 story pages read aloud. Falls back to Android TextToSpeech (en-US) or the browser's speech when a line has no clip.
Story data	`stories.js` (global `window.STORIES`)	30 stories with pages, questions, SWBST parts, main idea and model summary.
Word list	`words.js` (global `window.SR_WORDS` , 24,887 words)	Checks that a summary is written in real English words.
Storage	Browser `localStorage` (inside the WebView)	Profiles, progress, settings and the attempt log. See the Data Schema.

Android host	Java, one `MainActivity`, Android WebView	Full-screen portrait WebView, Back button, bridge `window.SummaRaceApp`.
Build	Gradle, Android Gradle Plugin 8.7.3, compileSdk / targetSdk 35, minSdk 24	Builds the APK. MP3 and WOFF2 files are stored uncompressed so the WebView can read them.
Version control	Git + GitHub (`orbitsdev/summaracev3`, public)	Source, pull requests and history.

4 Architecture



The dashed orange arrow is a lazy load: the 3D model files only load when the first race starts. No arrow leaves the phone: there is no server.

Script load order

`three.min.js` → `GLTFLoader.js` → `assets.js` → `stories.js` → `words.js` → `voice/voice.js` → the main script inside `index.html`. The main script is wrapped in one function (an IIFE), so its internals are not global. Each data file sets one global: `SR_ASSETS`, `STORIES`, `SR_WORDS`, `SR_VOICE`, `SR_MODELS`.

Main code modules inside index.html

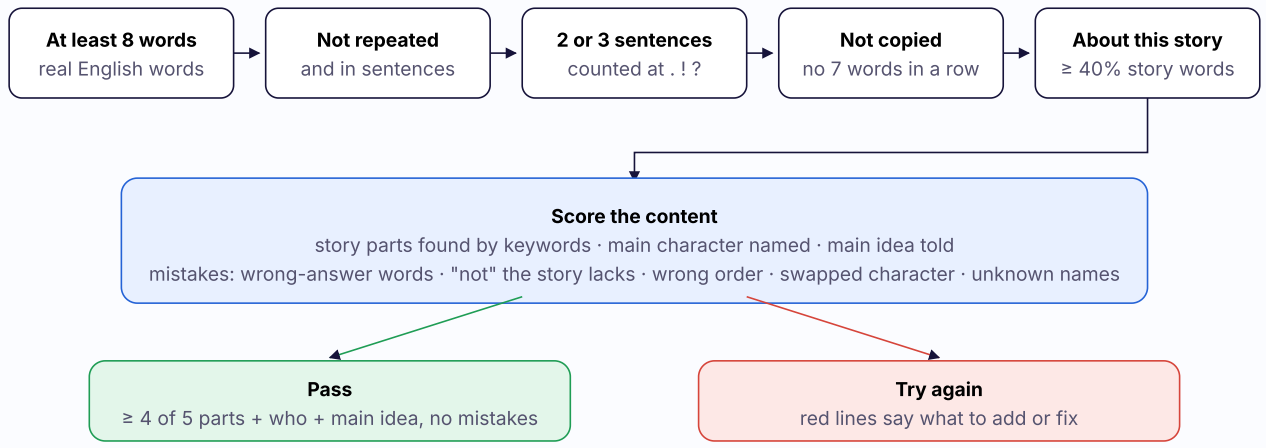
Module	What it does
Storage	`loadDb()`, `persist()`, preferences. One JSON object under `summarace.v1`.
Teacher Mira	`say()` shows a line and speaks it. Picks a recorded clip by normalised text (`vnorm`), maps lines with names or numbers to a recorded version, avoids repeating the same line, and folds her bubble when it covers a button.
Audio	Sound effects on Web Audio, music on looping `Audio` elements, voice with two priorities (story reading is never cut off by a short line).
Screens	`show(id)` switches sections: learner, home, read, quiz, order, race, summary, result, records.

Level flow	`startLevel` → `renderPage` / `askPage` → `askMain` → `startRace` → `startOrder` → `startSummary` → `finishLevel` .
Summary checker	Offline rules, see section 6.
Race	A self-contained module (`Race.prepare` , `Race.stop`) with the three.js scene, worlds, characters, obstacles, gates, patrol and end scenes.
Records	Table, CSV export (copy, download, share), erase with undo.

5 Race engine

- Three lanes at x = −2.2, 0 and 2.2. The runner (Max or Mia, chosen on the start card) is chased by an officer and a dog.
- A **gap** meter (0–100) is the distance to the patrol. It starts at 60 and shrinks slowly while running, but not while the learner is reading a question. A right answer adds 24. A wrong answer, a train bump or a barrier hit takes 10–14 away, but these never push the gap below 8: only the slow drain can end the race. At 0 the runner is caught and the race restarts.
- Each of the 5 SWBST parts is asked in turn. The question and the 3 answers show at the top. The game waits a reading time of 0.35 s per word (4–9 s, 25% less when a missed question comes back), then sends an answer gate down the track. The lane the runner is in at the gate is the answer.
- Each level has its own seeded course (`levelPlan(id)`): speed, obstacle rows and spacing grow with the level number, so level 30 is harder than level 1 but each level plays the same every time.
- Controls: swipe left/right to change lane, up to jump, down to roll. Tapping the left or right third of the screen also changes lane. Keyboard arrows (or A and D) and Space work in a desktop browser.
- Race score for the level = share of the 5 parts answered right on the first try, kept across "Try the race again".

6 Summary checker (offline)



A failed early check (top row) does not count as a try. From the 3rd scored try, mistakes no longer block a pass, and Mira shows her example. From the 4th, "Finish, my teacher will check" appears.

The checker is rules-based on purpose: it has to work offline on a cheap phone. It can miss things a teacher would notice, which is why the summary text and its status go into the CSV for the teacher to read.

7 Performance and device rules

- Start-up was measured at about 0.75 s with the CPU slowed 6× in a desktop browser. The 3D models (4.5 MB) are not part of start-up.
- Obstacles and effects free their GPU geometry and materials when removed; shared models and textures are kept.
- Portrait only. The layout uses safe-area insets for phones with a notch.
- Hardware acceleration is on for the WebView.

8 Build, run and release

```
Run in a browser:
  cd app/src/main/assets/www
  python -m http.server 8000      # then open http://localhost:8000

Build the APK:
  open the repo in Android Studio and build, or run: gradle assembleDebug
  the APK lands in app/build/outputs/apk/
```

- Raise `versionCode` and `versionName` in `app/build.gradle` for every release. The next free versionCode is 18.
- Release APKs are signed with the project's current signing key, which is kept off GitHub on the owner's PC. The original 1.2.0 key was lost, so phones with that old test build must uninstall it once.
- Never commit the keystore, `*.jks`, `*.keystore` or `local.properties`. They are in `.gitignore`.

9 Testing

- Day-to-day testing is in a desktop browser at `localhost:8000`, with the phone size set in the browser's device mode.
- Automated checks use Playwright with headless Chromium. Because the main code is private inside one function, tests reach it by injecting a small helper before the script ends.
- An APK is built for a phone test only when a change is ready. Android-only features (voice fallback, Download folder on Android 7–9, Back button) must be checked on a real phone.

PART 3 · APP FLOW

App Flow Document

SummaRace v3 · version 1.3.6

1 Overview

This document shows what happens when someone taps, and which screen comes next. It maps the main journeys through the game, from opening the app to finishing a level and exporting records.

2 Key user journeys

First play

Open app → PLAY → type name and section → World map → Level 1 → read, race, order, summarise → Results → Next.

Returning learner

PLAY → tap your name under "Or continue as" → map scrolls to your open level → Play.

Replay for more stars

Map → tap a finished level → Play again. Best stars and best score are kept.

Teacher collects results

PLAY → Teacher records (or Settings → Teacher records) → Copy, Download or Share CSV.

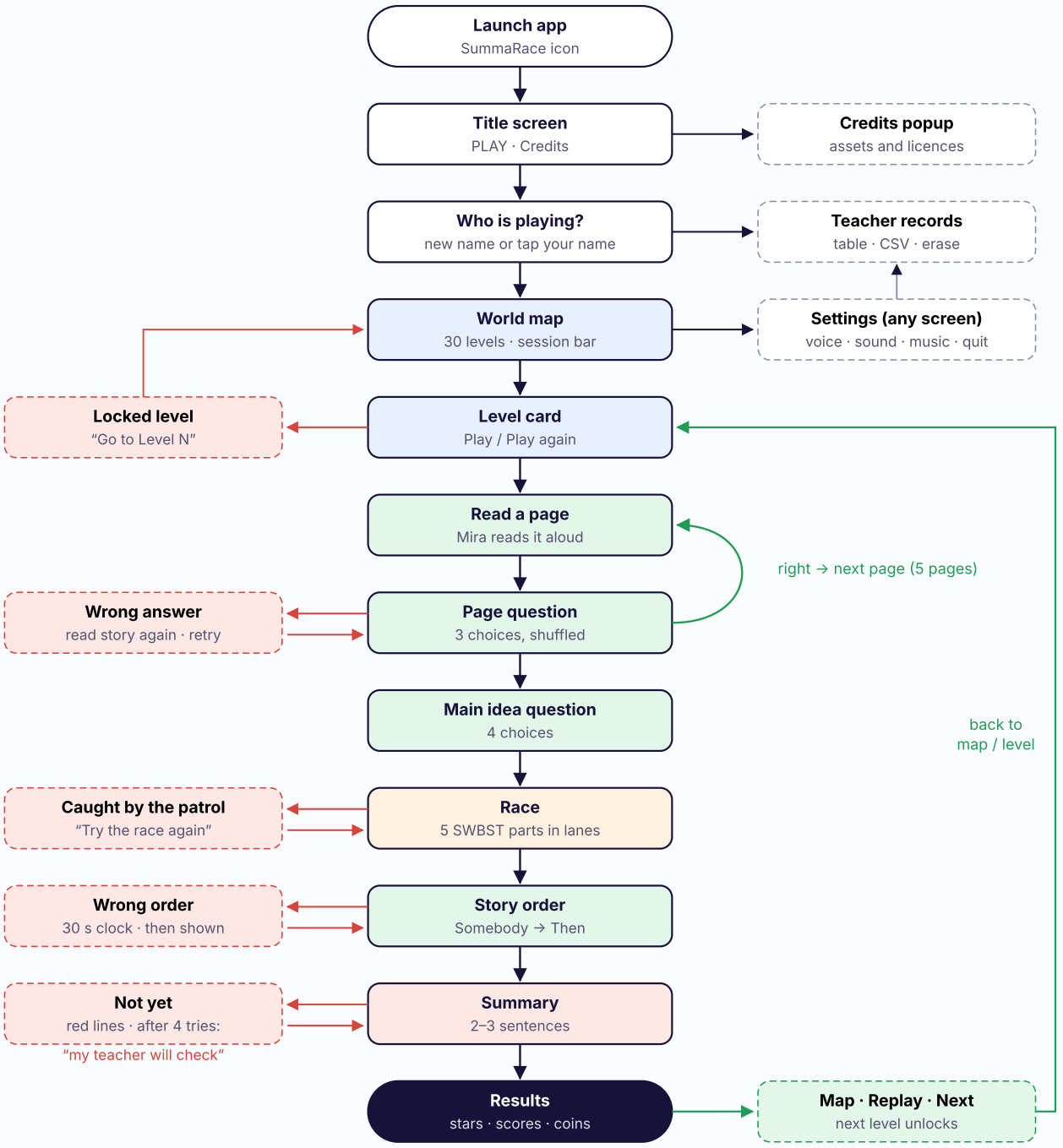
Stop mid-level

Settings or the Back button → Quit this story → World map. The unfinished level is not recorded.

Change player

Settings → Change player (or tap the name chip) → "Who is playing?".

3 App flow diagram



Solid boxes are the main path. Dashed boxes are side screens and retry loops. Green arrows move forward, red arrows are a retry or a block.

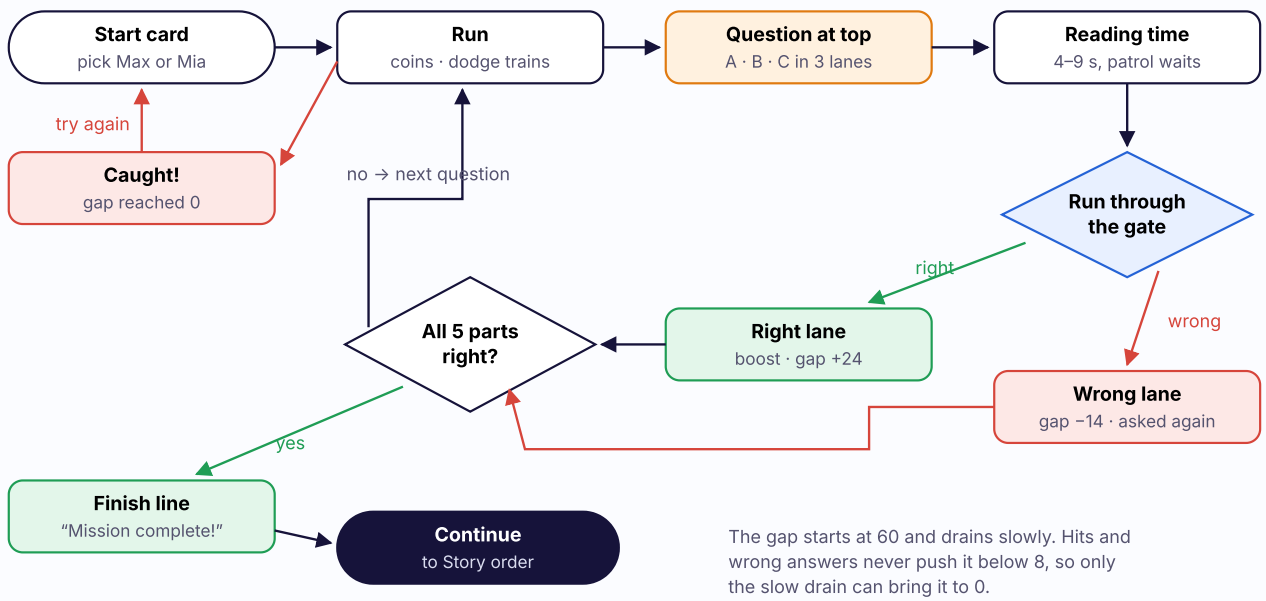
4 Inside one level

A level always runs in the same four steps, shown at the top of each screen as a step bar: **Read & answer → Race → Story order → Summary**.

Step	Screen	What the learner does	What happens next
1	Read	Reads page 1 of 5 (Mira reads it aloud too). Taps "Answer the question".	That page's question.

1	Page question	Picks 1 of 3 answers.	Right: "Next page" (or "Next: the main idea" after page 5). Wrong: Mira offers to show the story; after 2 wrong, the story opens with the part highlighted in yellow.
1	Main idea	Picks 1 of 4 answers.	Right: "Next: the race!".
2	Race	Start card → Start running. Answers 5 story-part questions by lane.	Mission complete → Continue.
3	Story order	Taps the 5 parts into Somebody, Wanted, But, So, Then. Taps "Check order".	All right: goes on by itself. Some wrong: green parts stay, 30-second clock, try again.
4	Summary	Writes 2–3 sentences. Taps "Check my summary".	Pass: "See my results". Not yet: feedback lines, try again.
—	Results	Sees stars and scores.	Map, Replay, or Next (hidden after level 30).

5 Race flow



The race is the only place a learner can "lose", and losing only restarts the race. The level score keeps first-try answers from every attempt.

6 Unlock rules

- A new profile starts with Level 1 open.
- Finishing level *n* (any score) opens level *n* + 1. Levels open strictly one by one.
- The session bar on the map groups levels in threes (session 1 = levels 1–3). A session chip is locked until its first level is open.
- Level *n* belongs to world ⌊(n − 1) ÷ 6⌋: levels 1–6 Sunny Meadow, 7–12 River Bend, 13–18 Story Town, 19–24 Palm Beach, 25–30 Summit Peak. The world sets the race scenery and the screen backdrop.

7 Back button

Where	Android Back does
Any popup open	Closes the popup.
Race, running	Pauses the race.
Race, any card showing	Opens the Quit popup.
Read, question, order, summary, results	Opens the Quit popup (Quit this story / Exit game), so a mistap never loses the level.
World map or records	Back to the title screen.
Title screen	Android's normal Back (leaves the app).

PART 4 · DESIGN BRIEF

Design Brief Document

SummaRace v3 · version 1.3.6 · screenshots taken from `main` at phone size (390 × 844)

1 Overview

This document sets the look of SummaRace: colours, fonts, components, and how each screen should look, so new screens match the ones already built.

2 Brand identity

Bright, friendly, readable. A cartoon "sticker" world: white and cream cards with thick plum outlines on a bright sky. Big rounded buttons that look pressable. Teacher Mira is always nearby, so the learner never feels alone with a hard question.

For 9–10 year olds

Large text, large tap targets (answers are at least 58 px tall), one task per screen.

Reading first

Story text uses Andika, a font made for early readers. Game chrome never covers the story.

Kind feedback

Wrong answers shake and turn red, but the right answer is never given away. Mira always offers to show the story again.

Game energy

Stars, coins, medals, confetti and a chase scene make finishing a level feel like a win.

3 Colour palette

Core

Plum
#3a1460 · outlines, titles

Ink
#2c1a4a · body text

Cream
#fffaf0 · cards

Sky
#2f8cf0 · background

Sun
#ffd23f · stars, highlights

Go
#22c06c · main buttons, right

Bad
#ff4a3d · wrong, close

Story parts (SWBST)

Each story part keeps one colour everywhere: the race, story order and summary feedback.

Somebody
#ff7a1a

Wanted
#ff4f9a

But
#e8352a

So
#1f8fe0

Then
#21b33c

Difficulty

Easy
#1fae4b

Average
#ff9300

Hard
#e8463b

The game has one bright look. It does not switch to a dark theme.

4 Typography

Role	Font	Used for
Title	Luckiest Guy	Logo, "AMAZING!" / "GREAT JOB!", flash words in the race, score numbers, the story's first letter.
Interface	Fredoka (variable, 300–700)	Buttons, labels, tags, step bar, popups.
Reading	Andika (regular, bold)	Story pages, questions, answers, the summary box.
Back-up	Lilita One, Atkinson Hyperlegible	Listed second in the font stacks; shown only if the main font can't load.

Headings are white with a thick plum outline on the sky, and plum on cards. Answer text is about 18 px (1.12 rem).

5 UI components

Component	Look	Use
Main button (`.btn.go`)	Green, rounded, white text, darker bottom edge that presses down 4 px on tap.	The one next step on a screen: Let's go!, Answer the question, Check order.
Second button (`.btn.alt`)	Purple or white with plum outline.	Back, Read the story, Download CSV.
Round icon button (`.cbtn`)	Circle with an icon; red, green, blue, yellow variants.	Home, Settings, Close (red ×), Listen (yellow), Map / Replay / Next.
Answer option (`.opt`)	White card with a coloured letter badge A/B/C/D. Turns green and pops when right, red and shakes when wrong.	Page questions, main idea.

Race lane chips	Three coloured chips (blue, orange, purple) with A/B/C, one per lane, under the question.	Race questions.
Slot (`.slot`)	Dashed box with a SWBST colour label; solid when filled, green with ✓ when right.	Story order.
Feedback row (`.fb row`)	Green "✓ Somebody" rows and red "Add: But" rows.	Summary check results.
Popup (`.modal .panel`)	Cream panel with a purple ribbon title and a red close button.	Who is playing?, Settings, Credits, Read it again.
Teacher Mira dock	Mira's picture at the bottom left with a speech bubble; folds to just her face.	Every screen except the race and records.
Step bar (`.steps`)	4 numbered dots: Read & answer, Race, Story order, Summary. Done steps show ✓.	Top of every level screen.

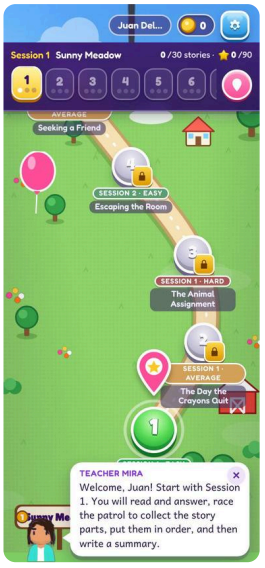
6 Screens



Title



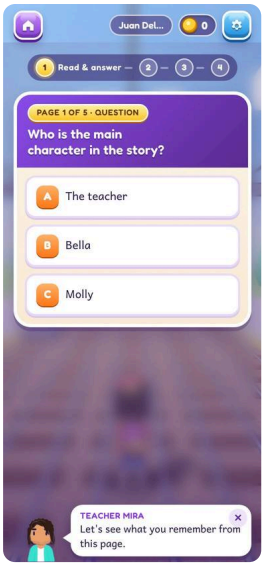
Who is playing?



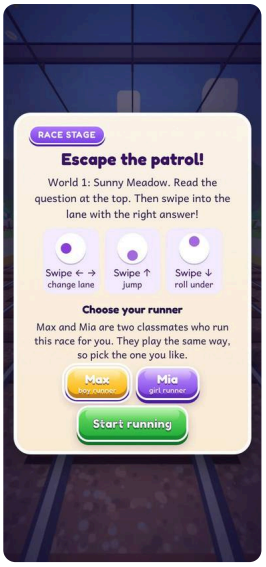
World map



Read a page



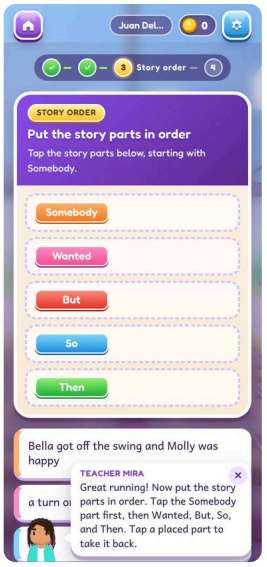
Page question



Race start card



Race: answers in lanes



Story order



Summary feedback



Results



Teacher records

Screenshots were taken in headless Chromium, so the 3D race uses software graphics and looks a little softer than on a phone.

7 Worlds

World	Levels	Race scenery	Screen backdrop
Sunny Meadow	1–6	Grass, flowers, houses, trees	`ui/world-0.jpg`
River Bend	7–12	River, houses, pines, mist	`ui/world-1.jpg`
Story Town	13–18	City buildings, trees	`ui/world-2.jpg`
Palm Beach	19–24	Sand, sea, palms, umbrellas	`ui/world-3.jpg`

Summit Peak	25–30	Snow, mountains, pines	`ui/world-4.jpg`

The race sky cycles through 6 times of day (morning to sunset) by level, so neighbouring levels look different.

8 Motion and sound

- Buttons press down on tap. Right answers pop, wrong answers shake. Screens slide in. Confetti on the results screen for 4.5 s.
- If the phone asks for reduced motion, all animations and confetti are turned off.
- Short sound effects for taps, coins, right, wrong, jump, bump and win. Music lowers while Mira talks.
- Voice, sound effects and music each have their own switch in Settings.

9 Accessibility

- Every icon button has a text label for screen readers. Mira's bubble is a live region, so her lines are read out.
- Focus outlines are a thick pink ring.
- Answer order is shuffled, so colour or position never tells the answer.
- Every story page can be read aloud with the Listen button.

PART 5 · DATA SCHEMA

Data Schema Document

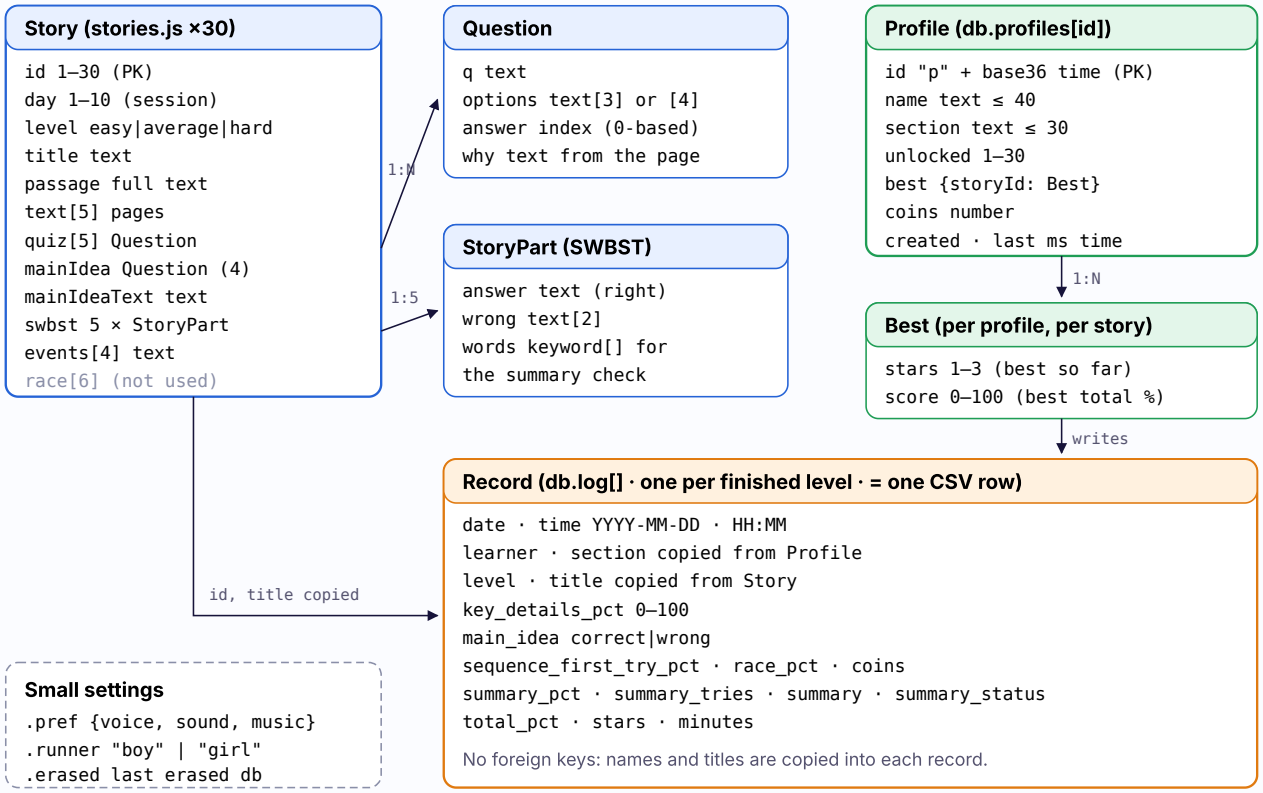
SummaRace v3 · version 1.3.6 · this replaces a "backend schema": the game has no backend

1 Overview

This document says where SummaRace's data lives, how it is structured, who can see it, and how the pieces connect.

> **There is no server and no database.** Story content is a JavaScript file shipped in the app. Learner progress and results are saved in the phone's WebView storage (`localStorage`). The only way data leaves the phone is when the teacher copies, downloads or shares the CSV.

2 Data model



Blue = shipped content (read-only). Green and orange = learner data saved on the phone. Monospace lines are field name and type.

3 Story content: stories.js

`window.STORIES` is an array of 30 objects, sorted by `id` when the game starts. Every story has 5 pages, 5 page questions with 3 options, a main idea question with 4 options, and 5 SWBST parts with 1 right and 2 wrong answers.

Field	Type	Used for
`id`	number 1–30	Level number and unlock order.
`day`	number 1–10	Session number shown on screens ("Session 1 · Easy").
`level`	`"easy" | "average" | "hard"`	Difficulty tag and colour. Each session has one of each.
`title`	string	Shown everywhere and copied into each record.
`passage`	string	The whole story in one piece. Used by the summary checker (copy check, names, vocabulary).
`text`	string[5]	The 5 pages the learner reads.
`quiz`	Question[5]	One question after each page. `why` is the page text that holds the answer; it is used to highlight the right part when the story is shown again.
`mainIdea`	Question (4 options)	The main idea question after page 5, and the main idea check in the summary.
`mainIdeaText`	string	The main idea in words; allowed vocabulary for the summary.
`swbst`	{somebody, wanted, but, so, then}: StoryPart	Race questions, story order items and summary scoring. `words` are the keywords that prove a summary told that part.
`events`	string[4]	Allowed vocabulary for the summary checker.
`modelSummary`	string	Teacher Mira's example, shown after the 3rd summary try and after a pass.
`race`	Question[6]	**Not used since 1.3.5.** The old trivia race questions. The race now asks the 5 SWBST parts instead.

```
{
  "id": 1, "day": 1, "level": "easy", "title": "The Playground",
  "text": ["Molly and her friend Bella were playing at the playground. …", "…"],
  "quiz": [{ "q": "Who is the main character in the story?",
             "options": ["Bella", "Molly", "The teacher"], "answer": 1, "why": "…"
  "swbst": { "somebody": { "answer": "Molly", "wrong": ["Bella", "The teacher"],
                           "words": ["molly", "girl", "kid"] }, … },
  "modelSummary": "Molly wanted her turn on the swing, but Bella would not get off.
}
```

Other shipped data files

File	Global	Shape
`words.js`	`SR_WORDS`	One string of 24,887 common English words separated by spaces.
`voice/voice.js`	`SR_VOICE`	Map from normalised line text (lower case, letters and digits only) to a clip path, e.g. `"please type your name first" → "voice/m-001.mp3"`. Story pages use the key `"page <storyId> <page>"`. 170 Mira lines + 150 pages = 320 clips.
`assets.js`	`SR_ASSETS`	`{textures:{platformer}, models:{coin, cloud}, sounds:{bad, bump, click, coin, good, hit, jump, land, swish, tap, win}}` as data URLs / base64.
`models/*.js`	`SR_MODELS`	`SR_MODELS[name] = "<base64 GLB>"` for boy, girl, officer, city, suburb, trains, nature.

4 Saved data: localStorage

Key	Value	Written when
`summarace.v1`	`{profiles: {id: Profile}, current: id | null, log: Record[]}`	A learner signs in, finishes a level, or the teacher erases or restores.
`summarace.v1.pref`	`{voice: bool, sound: bool, music: bool}` (all true by default)	A switch in Settings changes.
`summarace.v1.runner`	`"boy"` (Max) or `"girl"` (Mia)	The runner is picked on the race start card.
`summarace.v1.erased`	A full copy of `summarace.v1` from just before the last erase	The teacher types ERASE. Removed again by "Bring back erased records", which merges it with anything recorded since.

An unfinished level is never saved: quitting mid-level leaves no record. If storage is blocked or full, the game keeps working for the session but nothing is kept.

5 CSV export format

File name `summarace-records-YYYY-MM-DD.csv`, UTF-8 with a byte-order mark (so Excel opens it correctly), comma separated, one header row, then one row per finished level, oldest first. Values with a comma, quote or line break are wrapped in double quotes. The column order is fixed:

#	Column	Type	Meaning
1	`date`	YYYY-MM-DD	Date the level was finished (UTC date).
2	`time`	HH:MM	Local time on the phone.
3	`learner`	text	Name as typed.
4	`section`	text	Section or group as typed (may be empty).
5	`level`	1–30	Story id.
6	`title`	text	Story title.
7	`key_details_pct`	0–100	Page questions right on the first try.
8	`main_idea`	correct | wrong	Main idea right on the first try.
9	`sequence_first_try_pct`	0–100	Story parts in the right slot on the first "Check order".
10	`race_pct`	0–100	Story parts answered right on the first try in the race.
11	`coins`	number	Coins picked up in the race.
12	`summary_pct`	0–100	Story parts found in the passing summary, out of 5, minus half a part (10 points) for each wrong detail. 0 if handed to the teacher.
13	`summary_tries`	number	Scored summary checks (early checks like "too short" don't count).
14	`total_pct`	0–100	Average of read & answer, race, order and summary.
15	`stars`	1–3	Stars for this attempt (≥85 → 3, ≥65 → 2, else 1).
16	`minutes`	number, 1 decimal	Time from opening the level to the result.
17	`summary`	text	The learner's summary, word for word.

18	`summary_status`	passed | passed after example | teacher check	How the summary step ended. Added in 1.3.2, kept as the last column so old sheets still line up.

```
date,time,learner,section,level,title,key_details_pct,main_idea,sequence_first_try_
2026-11-05,08:06,Juan Dela Cruz,Grade 4 - Sampaguita,1,The Playground,100,correct,8
```

> Column `story_elements_first_try_pct` existed before 1.3.3 and was a copy of `race_pct`. It was removed, so sheets from older builds have one more column.

6 Access control

Who	Story content	Own progress	Other learners	Records / CSV
Learner	Read	Read, add by playing	Sees names in "Or continue as"	Can open Teacher records (no password)
Teacher	Read	—	Sees all on the phone	Copy, download, share, erase (type ERASE), restore
Anyone with the APK	Read (the files are inside the app; the repo is public)	Nothing: data stays on each phone		

There is no password on Teacher records. This was a choice for a classroom tool; the ERASE word and the undo protect against accidents, not against a learner who wants to look.

7 Data storage

Where

WebView localStorage inside the app's private folder. Clearing the app's data or uninstalling deletes it.

Export

Android 10+: Download/SummaRace via MediaStore. Android 7–9: Download/SummaRace with storage permission. Or Share / Copy.

Backups

None automatic. The teacher should export the CSV after each session.

Personal data

Only the name and section typed in, and the summaries. Nothing is sent anywhere.

PART 6 · IMPLEMENTATION PLAN

Implementation Plan Document

SummaRace v3 · written after the build, as a record of what was built and in what order

1 Overview

This document breaks SummaRace v3 into the phases it was actually built in, with what each phase delivered. Dates from 2026-10-27 on come from the git history. Earlier dates were not recorded and are marked "not recorded".

2 Goals

Deliver an offline Android reading game for the research team's 30 stories, with records the teacher can export, built and tested in small steps so each client revision could be checked on a phone before it was merged.

3 Timeline



The client revisions kept versionName 1.3.0 on main (a 1.3.1 test APK of that period is in the project files). 1.3.4 was a local test build only (PR #3 plus PR #5) and was never merged.

4 Implementation phases

Phase	What got built	Key deliverables	When
0. Earlier versions	SummaRace v1 and SummaRace2 in Unity.	Separate repos (`orbitsdev/SummaRace` , `orbitsdev/SummaRace2`). v3 is a rewrite, not a port.	Not recorded
1. Web game v3	The game as an HTML/JS app with three.js: title, profiles, world map, read and quiz, story order, SWBST, summary, results, records, 3D race, Teacher Mira's recorded voice, 30 stories.	SummaRace 1.2.0 test APK.	Not recorded
2. Recovery	The original project folder was lost. The web game was taken back out of the 1.2.0 APK and a new Android wrapper was written.	Repo created with the game in `app/src/main/assets/www` , `MainActivity.java` with the bridge, new signing key, 1.3.0. Lost for good: `build-voice.sh` , `CREDITS.md` , the old key.	2026-10-16 to 2026-10-27

3. Client final revisions	The client's revision list.	Race questions at a "Story stop" (later replaced), runner picker explained, one story-order instruction with a 30 s limit after a wrong try, summary in 2–3 sentences, offline content checks for each story part, wrong details, unknown names, "not", order and swapped characters.	2026-10-27 · PR #1
4. Bug review fixes	A 19-point review of the game, top items fixed.	Shuffled answers, Back button safety, Mira bubble folding, race GPU clean-up, device voice fallback, ERASE + undo, copy-check fix, "my teacher will check" + `summary_status` column, softer race penalty, order timer by Check, viewport and font fixes. 1.3.2 (versionCode 13).	2026-10-28 · PR #2
5. Review fixes 2	Two more review items.	Dropped the duplicate CSV column, CSV saves to Download/SummaRace on Android 7–9. 1.3.3 (versionCode 14).	2026-10-29 · PR #3 (merged 2026-11-05)
6. Lane answers	The client's last request: the race no longer stops for a question.	Question at the top, answers A/B/C in the 3 lanes, run through the right lane's gate. Reading time before each gate. 1.3.5 (versionCode 16).	2026-11-04 · PR #6
7. Race UI polish	Make the race readable on a phone.	Slim question panel, solved count in the top bar, Mira hidden during questions, centred start card with a swipe tutorial, on-screen arrow buttons removed. 1.3.6 (versionCode 17).	2026-11-04 · PR #7
8. Documentation	These documents and the full asset list.	`docs/` : PRD, TRD, App Flow, Design Brief, Data Schema, this plan, Assets & Credits.	2026-11-05

5 How each phase was run



PR #6 and PR #7 were merged before a full phone test, at the owner's request. A 1.3.6 test APK was built for that test.

6 Milestones

Milestone	Description	Status
Source recovered	Game runs from the repo in a browser and builds as an APK.	Done 2026-10-27
Client revisions applied	All items from the client's final revision list.	Done 2026-10-27
Review fixes	Top items of the 19-point review.	Done 2026-10-28/29
Race answers in lanes	Client's last request.	Done 2026-11-04
Documentation	This set of documents.	In review

Release build 1.3.6	Signed with the project key on the owner's PC, tested on a phone, handed to the client.	Owner to confirm

7 What is left

- Decide on PR #4 (race music speeds up) and close PR #5 (no longer needed).
- Ask the client about the two stories titled "In Grandfather's Day" (levels 15 and 18).
- Record voice clips for the newer Mira lines, or keep reusing recorded lines, since the clip build script was lost.
- Phone checks still worth doing: a phone with a notch, a 2 GB phone, the on-screen keyboard over the summary box, and a slow reader in the race.
- Next release must use versionCode 18 or higher.

PART 7 · ASSETS & CREDITS

Assets and Credits

Every file the game ships, where it came from, who made it, and its license. Checked against `main` (1.3.6) on 2026-11-05.

1 Overview

This page lists every asset inside `app/src/main/assets/www` and the Android launcher icons. Where the source or license is not recorded anywhere (the original `CREDITS.md` was lost with the original project folder), it says **unknown** instead of guessing. One item is marked **likely** where the file itself points to a source but nothing records it.

3D models

7 files · 4.5 MB · Quaternius and Kenney, CC0

Voice clips

320 MP3 · 8.0 MB · made with Kokoro TTS

Music

2 MP3 · 2.2 MB · source unknown

Images

18 files · 0.9 MB · source unknown

Fonts

7 files · 160 KB · Google Fonts, OFL / Apache

Sounds

11 clips inside assets.js · Kenney, CC0

2 Gaps to fix before a public release

These are not guesses; each one is visible in the files.

- **Music and all UI images have no recorded source.** The credits screen calls the music "original loops, made for this game" but nobody has a record of who made it or with what tool. The files carry a C2PA tag saying they passed through Claude's file handling, which only shows how they were copied, not who made them. The owner should confirm where Mira, the world pictures, the backdrop, stars, icons and both music loops came from.
- **wordfreq (CC BY-SA 4.0) is not credited on the credits screen.** That license requires credit. Add "Word list: wordfreq by Robyn Speer (CC BY-SA 4.0)".
- **The credits screen says "Full list in CREDITS.md"**, but that file was lost and is not in the app. Point it to this page or ship a new CREDITS file.
- **The font license texts are not shipped.** The SIL OFL and Apache 2.0 both ask for the license text to travel with the font files. Add `OFL.txt` and the Apache license next to `fonts/`.
- **Stories have no author credit.** Several are published children's stories (for example "The Day the Crayons Quit" and "Owen and Mzee"). The research team should confirm they may be used and how to credit them.
- **Teacher Mira has no credit line** on the credits screen, and no repo LICENSE says who owns the game code.

3 3D models

Models

File	Size	Used in the game	Source	Author	License
`models/boy.js`	906 KB	Max, the boy runner (animations: Run, Jump, Roll, Idle, Victory, Defeat)	quaternius.com · Ultimate Animated Character Pack	Quaternius	CC0 1.0 (public domain) (as stated on the credits screen)
`models/girl.js`	911 KB	Mia, the girl runner (same animations)	Same pack	Quaternius	CC0 1.0 (public domain)
`models/officer.js`	951 KB	The patrol officer who chases the runner (Run, Idle, Victory)	Same pack	Quaternius	CC0 1.0 (public domain)
`models/trains.js`	155 KB	Subway trains on the track (train-electric-subway-a/b/c, wheels, connector)	Kenney Train Kit	Kenney (kenney.nl)	CC0 1.0 (public domain)
`models/city.js`	1001 KB	Buildings and skyscrapers beside the track (Story Town)	Kenney City Kit (Commercial)	Kenney (kenney.nl)	CC0 1.0 (public domain)
`models/suburb.js`	467 KB	Houses, trees, fences and planters (meadow and river worlds)	Kenney City Kit (Suburban)	Kenney (kenney.nl)	CC0 1.0 (public domain)
`models/nature.js`	182 KB	Trees, palms, pines, rocks, bushes, flowers, mushrooms, logs	Kenney Nature Kit	Kenney (kenney.nl)	CC0 1.0 (public domain)
`assets.js` → coin, cloud	45 KB	Coins on the track, clouds in the sky	Kenney, exact kit not recorded. likely Platformer Kit: the shared texture is named “platformer”	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → platformer texture	10 KB	Colour map for the coin and cloud models	Same as above	Kenney (kenney.nl)	CC0 1.0 (public domain)

Characters are recoloured and given cartoon shading in code. Mesh names in the files match the Kenney kits listed (for example `train-electric-subway-a`, `building-skyscraper-a`, `building-type-a`, `tree_palmTall`). All model files were compressed with glTF-Transform 4.1.1 and stored as base64.

4 Sound effects

Sounds inside assets.js

File	Size	Used in the game	Source	Author	License
`assets.js` → click	1 KB	Most button taps	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → tap	1 KB	Small taps	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → good	5 KB	Right answer	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → bad	5 KB	Wrong answer	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)

`assets.js` → win	2 KB	Level, race or summary passed	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → coin	3 KB	Coin picked up	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → jump	2 KB	Runner jumps	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → land	1 KB	Runner lands	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → swish	1 KB	Page turn, roll	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → bump	3 KB	Hit a train, caught	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → hit	7 KB	Hit a barrier	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
beep, pop	—	Countdown and small pops	Made in code with Web Audio (no file)	SummaRace code	—

5 Music

Music

File	Size	Used in the game	Source	Author	License
`music/menu.mp3`	1.0 MB	Background music on every screen except the race (about 90 s loop)	unknown The credits screen says “Original loops, made for this game”, but who made them and how is not recorded	unknown	unknown
`music/race.mp3`	1.1 MB	Race music (about 72 s loop, A minor, ~161 bpm)	unknown Same as above	unknown	unknown

> Not in the game: three Kevin MacLeod tracks (Pixel Peeker Polka – faster, Monkeys Spinning Monkeys, Run Amok, from incompetech.com, CC BY 4.0) were downloaded as previews only. If one is ever used, it needs the credit line "Music by Kevin MacLeod (incompetech.com), licensed under Creative Commons: By Attribution 4.0".

6 Voice: Teacher Mira

Item	Detail
Files	`voice/m-000.mp3` to `voice/m-319.mp3` : 320 clips, 8.0 MB. 170 are Mira's lines, 150 are the 30 stories × 5 pages read aloud.
Made with	Kokoro TTS, recorded offline (credits screen and `voice.js` header). Model: hexgrad/Kokoro-82M.
Author	Model by hexgrad. Which Kokoro voice was used: unknown
License	Apache License 2.0 (the model). The clips were generated for this game.
Build script	`build-voice.sh` made the clips and `voice.js` . It was lost, so new lines can't be recorded in the same voice yet.

Fallback	A line with no clip is spoken by the phone's own voice (Android TextToSpeech, en-US) or the browser's voice.

Lines that have no clip

All 80 lines in Mira's main line lists have clips. These fixed lines in the code do not, so they use the phone voice:

- Time's up! Here is the right order. Read it once, then let's write your summary.
- Last stage! Write 2 or 3 sentences that tell the story in your own words. Before you check, ask yourself: ...
- Tell me more! Write 2 or 3 sentences: who the story is about, what happened, and how it ended.
- Some words are repeated too many times. Tell the story in sentences that make sense.
- Write it in sentences, not a list of words. Use words like wanted, but, so, and then.
- I see only one sentence. Please write 2 or 3 sentences, with a period at the end of each one.
- Some details don't match the story. Look at the red lines and fix them.
- Who is the story about? Name the main character in your summary.
- Your summary needs the main idea. What is the whole story mostly about?
- Good effort! Your teacher will read your summary and help you with it.

Lines built from a template (with a name, number or story title) were not checked one by one and may also fall back.

▼ **Full list of all 320 voice clips** (line text is stored in lower case without punctuation)

File	Kind	Line / page	Size
`m-000.mp3`	Mira line	hi i m teacher mira type your name so i can save your progress then we ll start reading together	28.8 KB
`m-001.mp3`	Mira line	please type your name first	12.3 KB
`m-002.mp3`	Mira line	that level is still locked finish the open level first and it will unlock	23.1 KB
`m-003.mp3`	Mira line	welcome start with level 1 you will read answer put events in order race the patrol and then summarize	32.1 KB
`m-004.mp3`	Mira line	you finished all 30 stories replay any level to earn more stars	22.1 KB
`m-005.mp3`	Mira line	welcome back your next level is ready for you	16.4 KB
`m-006.mp3`	Mira line	read carefully look for somebody what they wanted the problem what they did and how it ended	28.1 KB
`m-007.mp3`	Mira line	take your time think who is the story about and what is the big idea	23.1 KB
`m-008.mp3`	Mira line	picture the story in your head it helps you remember what happened first next then and last	27.1 KB
`m-009.mp3`	Mira line	let s check the key details choose the best answer	17.7 KB
`m-010.mp3`	Mira line	now the big one the main idea which answer tells what the whole story is about not just one small part	32.2 KB
`m-011.mp3`	Mira line	correct great memory	12.0 KB
`m-012.mp3`	Mira line	yes that s right	10.6 KB

m-013.mp3	Mira line	exactly you read closely	13.2 KB
m-014.mp3	Mira line	not quite the main idea covers the whole story look at the green answer	23.5 KB
m-015.mp3	Mira line	not quite look at the green answer and the reason below	19.4 KB
m-016.mp3	Mira line	put the story in order tap the event that happened first then next then then and last tap a placed event to take it back	35.6 KB
m-017.mp3	Mira line	perfect order on the first try now the race pick the lanes with the right answers to escape the patrol	31.7 KB
m-018.mp3	Mira line	all in order now time for the race pick the right answer lanes to escape	23.7 KB
m-019.mp3	Mira line	you escaped now find the story elements choose the best answer for somebody wanted but so and then	31.3 KB
m-020.mp3	Mira line	all five story elements are correct now use them to write your summary	22.8 KB
m-021.mp3	Mira line	now all five are correct let s use them to write your summary	20.3 KB
m-022.mp3	Mira line	last stage join the story elements into one sentence using your own words use words like wanted but so and then	36.0 KB
m-023.mp3	Mira line	i see more than one sentence join your ideas into one sentence with words like but so and then	30.4 KB
m-024.mp3	Mira line	that sentence is very long a summary keeps only the main idea try to use fewer words	28.7 KB
m-025.mp3	Mira line	part of this is copied straight from the story try saying it in your own words	23.4 KB
m-026.mp3	Mira line	excellent your sentence has every story element	17.1 KB
m-027.mp3	Mira line	the red chips show what is still missing add them in your own words	20.3 KB
m-028.mp3	Mira line	you completed the final story i m so proud of you	17.8 KB
m-029.mp3	Mira line	replay this level anytime to earn more stars	17.8 KB
m-030.mp3	Mira line	three stars you re a true summaracer	14.4 KB
m-031.mp3	Mira line	the next level is now unlocked replay this level anytime to earn more stars	25.2 KB
m-032.mp3	Mira line	the next level is now unlocked three stars you re a true summaracer	23.2 KB
m-033.mp3	Mira line	read the question then pick your lane	14.4 KB
m-034.mp3	Mira line	which lane has the right answer	12.8 KB

m-035.mp3	Miraline	think about the story then move	13.5 KB
m-036.mp3	Miraline	watch out for trains	11.2 KB
m-037.mp3	Miraline	yes you remembered	11.2 KB
m-038.mp3	Miraline	great answer keep going	12.3 KB
m-039.mp3	Miraline	correct the patrol is falling behind	15.2 KB
m-040.mp3	Miraline	oops that was not the right answer the patrol is closing in	20.0 KB
m-041.mp3	Miraline	trains block the lane switch lanes early	16.5 KB
m-042.mp3	Miraline	jump over the low barriers or switch lanes	15.4 KB
m-043.mp3	Miraline	swipe down to roll under the high barriers	15.3 KB
m-044.mp3	Miraline	the patrol is right behind you pick the correct answer lanes to get away	22.8 KB
m-045.mp3	Miraline	your device can t read aloud but you can read it yourself	19.1 KB
m-046.mp3	Miraline	0 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	35.2 KB
m-047.mp3	Miraline	1 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	34.3 KB
m-048.mp3	Miraline	2 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	34.5 KB
m-049.mp3	Miraline	3 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	34.4 KB
m-050.mp3	Miraline	0 of 5 are correct change the red ones and check again	20.9 KB
m-051.mp3	Miraline	1 of 5 are correct change the red ones and check again	19.9 KB
m-052.mp3	Miraline	2 of 5 are correct change the red ones and check again	19.9 KB
m-053.mp3	Miraline	3 of 5 are correct change the red ones and check again	19.8 KB
m-054.mp3	Miraline	4 of 5 are correct change the red ones and check again	20.1 KB
m-055.mp3	Miraline	that s only 1 word a good one sentence summary has about 15 to 30 words	26.1 KB
m-056.mp3	Miraline	that s only 2 words a good one sentence summary has about 15 to 30 words	26.2 KB

m-057.mp3	Mira line	that s only 3 words a good one sentence summary has about 15 to 30 words	26.8 KB
m-058.mp3	Mira line	that s only 4 words a good one sentence summary has about 15 to 30 words	26.6 KB
m-059.mp3	Mira line	that s only 5 words a good one sentence summary has about 15 to 30 words	27.0 KB
m-060.mp3	Mira line	that s only 6 words a good one sentence summary has about 15 to 30 words	26.5 KB
m-061.mp3	Mira line	that s only 7 words a good one sentence summary has about 15 to 30 words	27.1 KB
m-062.mp3	Mira line	that s only 8 words a good one sentence summary has about 15 to 30 words	26.2 KB
m-063.mp3	Mira line	that s only 9 words a good one sentence summary has about 15 to 30 words	26.9 KB
m-064.mp3	Mira line	good summary you included 3 of 5 story elements	18.6 KB
m-065.mp3	Mira line	good summary you included 4 of 5 story elements	18.7 KB
m-066.mp3	Mira line	you included 0 of 5 story elements look at the story elements above and add the missing parts	30.6 KB
m-067.mp3	Mira line	you included 1 of 5 story elements look at the story elements above and add the missing parts	29.3 KB
m-068.mp3	Mira line	you included 2 of 5 story elements look at the story elements above and add the missing parts	29.8 KB
m-069.mp3	Mira line	you included 3 of 5 story elements look at the story elements above and add the missing parts	29.4 KB
m-070.mp3	Mira line	you included 4 of 5 story elements look at the story elements above and add the missing parts	29.9 KB
m-071.mp3	Mira line	hi i m teacher mira tap play then type your name so i can save your progress	25.8 KB
m-072.mp3	Mira line	type your name or tap your name if you played before	17.4 KB
m-073.mp3	Mira line	let s read together who is this story about what do they want	19.9 KB
m-074.mp3	Mira line	what problem happens keep reading carefully	16.5 KB
m-075.mp3	Mira line	how does the story end remember what happened first next then and last	22.9 KB
m-076.mp3	Mira line	here is the story again read the yellow part carefully then try again	23.1 KB
m-077.mp3	Mira line	here is the story again read carefully then try again	20.0 KB
m-078.mp3	Mira line	not quite do you want to read the story again	15.8 KB

m-079.mp3	Mira line	choose another answer you can do it	14.3 KB
m-080.mp3	Mira line	some events are not in the right place yet the green ones stay do you want to read the story again	27.9 KB
m-081.mp3	Mira line	ask yourself what had to happen first	15.2 KB
m-082.mp3	Mira line	you escaped now find the story elements one at a time	19.6 KB
m-083.mp3	Mira line	i can t read some of these words please write your summary in english using real words	26.3 KB
m-084.mp3	Mira line	some words are repeated too many times tell the story in a sentence that makes sense	25.3 KB
m-085.mp3	Mira line	good summary you told the main idea in your own words	18.1 KB
m-086.mp3	Mira line	who is the story about name the main character in your sentence	20.5 KB
m-087.mp3	Mira line	your sentence is missing some story elements look at the story elements above and add the missing parts	30.7 KB
m-088.mp3	Mira line	your sentence needs the main idea what is the whole story mostly about	24.1 KB
m-089.mp3	Mira line	which card has the right answer	12.7 KB
m-090.mp3	Mira line	this question is back think about the story and try again	19.3 KB
m-091.mp3	Mira line	not quite that question will come back the patrol is closing in	20.9 KB
m-092.mp3	Mira line	you got them all run to the finish line	15.2 KB
m-093.mp3	Mira line	you made it mission complete	13.1 KB
m-094.mp3	Mira line	write it as a sentence not a list of words use words like wanted but so and then	26.7 KB
m-095.mp3	Mira line	you got it	9.4 KB
m-096.mp3	Mira line	that s it well done	11.1 KB
m-097.mp3	Mira line	nice work that s correct	12.1 KB
m-098.mp3	Mira line	super you remembered	11.6 KB
m-099.mp3	Mira line	right answer keep it up	12.3 KB
m-100.mp3	Mira line	wonderful you were paying attention	14.3 KB

m-101.mp3	Miraline	yes you found it in the story	12.9 KB
m-102.mp3	Miraline	great thinking	10.1 KB
m-103.mp3	Miraline	perfect on to the next one	12.5 KB
m-104.mp3	Miraline	awesome that s the one	11.9 KB
m-105.mp3	Miraline	way to go	9.5 KB
m-106.mp3	Miraline	you re a careful reader	11.0 KB
m-107.mp3	Miraline	good job that s correct	12.0 KB
m-108.mp3	Miraline	brilliant you know this story	12.4 KB
m-109.mp3	Miraline	yes indeed nicely done	12.9 KB
m-110.mp3	Miraline	hmm not that one want to look at the story again	17.3 KB
m-111.mp3	Miraline	almost let s check the story together	14.8 KB
m-112.mp3	Miraline	not yet the story can help you want to read it again	18.5 KB
m-113.mp3	Miraline	good try let s look back at the story	15.4 KB
m-114.mp3	Miraline	oops not that one shall we read that part again	17.0 KB
m-115.mp3	Miraline	that s not it yet take another look at the story	16.9 KB
m-116.mp3	Miraline	close but not quite want to peek at the story	16.9 KB
m-117.mp3	Miraline	try again think about what you read	14.5 KB
m-118.mp3	Miraline	take your time and try another one	14.1 KB
m-119.mp3	Miraline	you can do this pick again	12.7 KB
m-120.mp3	Miraline	read the question once more then choose	15.2 KB
m-121.mp3	Miraline	think about the story then try again	14.6 KB
m-122.mp3	Miraline	don t give up try a different answer	14.2 KB

m-123.mp3	Mira line	let s turn the page	11.0 KB
m-124.mp3	Mira line	here s the next page	11.1 KB
m-125.mp3	Mira line	on to the next page	11.1 KB
m-126.mp3	Mira line	let s keep reading	10.8 KB
m-127.mp3	Mira line	ready for more here s the next page	15.0 KB
m-128.mp3	Mira line	now a question about this page	13.2 KB
m-129.mp3	Mira line	let s see what you remember from this page	14.2 KB
m-130.mp3	Mira line	question time think about what you just read	16.4 KB
m-131.mp3	Mira line	here s a question about that page	13.5 KB
m-132.mp3	Mira line	here s a new story read carefully and look for the main character	20.6 KB
m-133.mp3	Mira line	story time who is in this story and what do they want	18.6 KB
m-134.mp3	Mira line	let s begin read slowly and picture the story in your mind	20.2 KB
m-135.mp3	Mira line	keep going watch for the problem in the story	16.4 KB
m-136.mp3	Mira line	what is happening now read carefully	14.7 KB
m-137.mp3	Mira line	think about what the character does next	14.1 KB
m-138.mp3	Mira line	which card has the right story part	14.0 KB
m-139.mp3	Mira line	find the right card and run into it	14.4 KB
m-140.mp3	Mira line	look at all three cards then choose	14.5 KB
m-141.mp3	Mira line	remember the story which one fits	14.5 KB
m-142.mp3	Mira line	boost that s the right part	12.6 KB
m-143.mp3	Mira line	nice catch keep running	12.1 KB
m-144.mp3	Mira line	you got it speed up	11.3 KB

m-145.mp3	Mira line	perfect the patrol can t keep up	14.4 KB
m-146.mp3	Mira line	not quite that one will come back the patrol is closing in	19.3 KB
m-147.mp3	Mira line	oops wrong card you ll see it again soon	15.9 KB
m-148.mp3	Mira line	not that one think about the story and try again later	19.0 KB
m-149.mp3	Mira line	careful that part will come back run faster	17.1 KB
m-150.mp3	Mira line	this one is back think about the story and try again	18.4 KB
m-151.mp3	Mira line	here it is again which card is right this time	16.8 KB
m-152.mp3	Mira line	second chance remember what happened in the story	17.0 KB
m-153.mp3	Mira line	all the story parts are yours race to the finish	17.2 KB
m-154.mp3	Mira line	that s all five now run to the finish line	16.2 KB
m-155.mp3	Mira line	finish line you escaped the patrol	14.2 KB
m-156.mp3	Mira line	hooray mission complete	12.2 KB
m-157.mp3	Mira line	this is the last page how does it all end	15.7 KB
m-158.mp3	Mira line	last page think about how the problem was solved	17.1 KB
m-159.mp3	Mira line	almost done read how the story ends	15.3 KB
m-160.mp3	Mira line	great running now put the story parts in order tap the somebody part first then wanted but so and then tap a placed part to take it back	41.4 KB
m-161.mp3	Mira line	perfect order on the first try now use these parts to write your summary	23.1 KB
m-162.mp3	Mira line	all in order now now use these parts to write your summary	20.0 KB
m-163.mp3	Mira line	some parts are not in the right place yet the green ones stay do you want to read the story again	27.6 KB
m-164.mp3	Mira line	ask yourself who is it about what did they want and what went wrong	20.6 KB
m-165.mp3	Mira line	the race could not start on this device so let s keep going	18.9 KB
m-166.mp3	Mira line	last stage write one sentence that tells the main idea of the story in your own words before you check ask yourself did i tell who what they wanted the problem what they did and how it ended	55.2 KB

`m-167.mp3`	Mira line	your sentence is missing some story parts did you tell who what they wanted the problem what they did and how it ended	34.6 KB
`m-168.mp3`	Mira line	the patrol is right behind you collect the right story parts to get away	23.1 KB
`m-169.mp3`	Mira line	welcome start with session 1 you will read and answer race the patrol to collect the story parts put them in order and then write a summary	39.8 KB
`page-01-1.mp3`	Story page	Level 1 “The Playground”, page 1	47.8 KB
`page-01-2.mp3`	Story page	Level 1 “The Playground”, page 2	44.1 KB
`page-01-3.mp3`	Story page	Level 1 “The Playground”, page 3	57.2 KB
`page-01-4.mp3`	Story page	Level 1 “The Playground”, page 4	73.2 KB
`page-01-5.mp3`	Story page	Level 1 “The Playground”, page 5	29.6 KB
`page-02-1.mp3`	Story page	Level 2 “The Day the Crayons Quit”, page 1	35.3 KB
`page-02-2.mp3`	Story page	Level 2 “The Day the Crayons Quit”, page 2	18.0 KB
`page-02-3.mp3`	Story page	Level 2 “The Day the Crayons Quit”, page 3	23.8 KB
`page-02-4.mp3`	Story page	Level 2 “The Day the Crayons Quit”, page 4	30.4 KB
`page-02-5.mp3`	Story page	Level 2 “The Day the Crayons Quit”, page 5	24.3 KB
`page-03-1.mp3`	Story page	Level 3 “The Animal Assignment”, page 1	29.9 KB
`page-03-2.mp3`	Story page	Level 3 “The Animal Assignment”, page 2	29.5 KB
`page-03-3.mp3`	Story page	Level 3 “The Animal Assignment”, page 3	24.6 KB
`page-03-4.mp3`	Story page	Level 3 “The Animal Assignment”, page 4	29.5 KB
`page-03-5.mp3`	Story page	Level 3 “The Animal Assignment”, page 5	34.3 KB
`page-04-1.mp3`	Story page	Level 4 “Escaping the Room”, page 1	24.8 KB
`page-04-2.mp3`	Story page	Level 4 “Escaping the Room”, page 2	22.0 KB
`page-04-3.mp3`	Story page	Level 4 “Escaping the Room”, page 3	19.4 KB
`page-04-4.mp3`	Story page	Level 4 “Escaping the Room”, page 4	27.1 KB

`page-04-5.mp3`	Story page	Level 4 “Escaping the Room”, page 5	24.5 KB
`page-05-1.mp3`	Story page	Level 5 “Seeking a Friend”, page 1	31.4 KB
`page-05-2.mp3`	Story page	Level 5 “Seeking a Friend”, page 2	19.7 KB
`page-05-3.mp3`	Story page	Level 5 “Seeking a Friend”, page 3	16.3 KB
`page-05-4.mp3`	Story page	Level 5 “Seeking a Friend”, page 4	22.0 KB
`page-05-5.mp3`	Story page	Level 5 “Seeking a Friend”, page 5	25.9 KB
`page-06-1.mp3`	Story page	Level 6 “Shawn the Speedy Snail”, page 1	36.2 KB
`page-06-2.mp3`	Story page	Level 6 “Shawn the Speedy Snail”, page 2	22.8 KB
`page-06-3.mp3`	Story page	Level 6 “Shawn the Speedy Snail”, page 3	30.2 KB
`page-06-4.mp3`	Story page	Level 6 “Shawn the Speedy Snail”, page 4	31.0 KB
`page-06-5.mp3`	Story page	Level 6 “Shawn the Speedy Snail”, page 5	30.9 KB
`page-07-1.mp3`	Story page	Level 7 “Emma's Favorite Restaurant”, page 1	32.6 KB
`page-07-2.mp3`	Story page	Level 7 “Emma's Favorite Restaurant”, page 2	22.0 KB
`page-07-3.mp3`	Story page	Level 7 “Emma's Favorite Restaurant”, page 3	21.5 KB
`page-07-4.mp3`	Story page	Level 7 “Emma's Favorite Restaurant”, page 4	21.8 KB
`page-07-5.mp3`	Story page	Level 7 “Emma's Favorite Restaurant”, page 5	25.9 KB
`page-08-1.mp3`	Story page	Level 8 “The Case of the Missing Lunch”, page 1	27.4 KB
`page-08-2.mp3`	Story page	Level 8 “The Case of the Missing Lunch”, page 2	19.9 KB
`page-08-3.mp3`	Story page	Level 8 “The Case of the Missing Lunch”, page 3	20.8 KB
`page-08-4.mp3`	Story page	Level 8 “The Case of the Missing Lunch”, page 4	26.4 KB
`page-08-5.mp3`	Story page	Level 8 “The Case of the Missing Lunch”, page 5	24.8 KB
`page-09-1.mp3`	Story page	Level 9 “The Selfish Giant”, page 1	27.6 KB

`page-09-2.mp3`	Story page	Level 9 “The Selfish Giant”, page 2	22.3 KB
`page-09-3.mp3`	Story page	Level 9 “The Selfish Giant”, page 3	20.1 KB
`page-09-4.mp3`	Story page	Level 9 “The Selfish Giant”, page 4	23.5 KB
`page-09-5.mp3`	Story page	Level 9 “The Selfish Giant”, page 5	17.8 KB
`page-10-1.mp3`	Story page	Level 10 “A Visit to the Zoo”, page 1	27.6 KB
`page-10-2.mp3`	Story page	Level 10 “A Visit to the Zoo”, page 2	14.5 KB
`page-10-3.mp3`	Story page	Level 10 “A Visit to the Zoo”, page 3	22.1 KB
`page-10-4.mp3`	Story page	Level 10 “A Visit to the Zoo”, page 4	17.2 KB
`page-10-5.mp3`	Story page	Level 10 “A Visit to the Zoo”, page 5	32.5 KB
`page-11-1.mp3`	Story page	Level 11 “Be Careful What You Wish For”, page 1	35.4 KB
`page-11-2.mp3`	Story page	Level 11 “Be Careful What You Wish For”, page 2	29.6 KB
`page-11-3.mp3`	Story page	Level 11 “Be Careful What You Wish For”, page 3	36.1 KB
`page-11-4.mp3`	Story page	Level 11 “Be Careful What You Wish For”, page 4	26.3 KB
`page-11-5.mp3`	Story page	Level 11 “Be Careful What You Wish For”, page 5	33.7 KB
`page-12-1.mp3`	Story page	Level 12 “AgitAgueda”, page 1	35.8 KB
`page-12-2.mp3`	Story page	Level 12 “AgitAgueda”, page 2	26.5 KB
`page-12-3.mp3`	Story page	Level 12 “AgitAgueda”, page 3	29.5 KB
`page-12-4.mp3`	Story page	Level 12 “AgitAgueda”, page 4	36.6 KB
`page-12-5.mp3`	Story page	Level 12 “AgitAgueda”, page 5	28.6 KB
`page-13-1.mp3`	Story page	Level 13 “The Crowded House: A Folktale”, page 1	38.1 KB
`page-13-2.mp3`	Story page	Level 13 “The Crowded House: A Folktale”, page 2	23.8 KB
`page-13-3.mp3`	Story page	Level 13 “The Crowded House: A Folktale”, page 3	34.7 KB

`page-13-4.mp3`	Story page	Level 13 “The Crowded House: A Folktale”, page 4	35.5 KB
`page-13-5.mp3`	Story page	Level 13 “The Crowded House: A Folktale”, page 5	32.4 KB
`page-14-1.mp3`	Story page	Level 14 “Owen and Mzee”, page 1	32.9 KB
`page-14-2.mp3`	Story page	Level 14 “Owen and Mzee”, page 2	20.6 KB
`page-14-3.mp3`	Story page	Level 14 “Owen and Mzee”, page 3	27.6 KB
`page-14-4.mp3`	Story page	Level 14 “Owen and Mzee”, page 4	20.7 KB
`page-14-5.mp3`	Story page	Level 14 “Owen and Mzee”, page 5	23.4 KB
`page-15-1.mp3`	Story page	Level 15 “In Grandfather's Day”, page 1	29.1 KB
`page-15-2.mp3`	Story page	Level 15 “In Grandfather's Day”, page 2	23.8 KB
`page-15-3.mp3`	Story page	Level 15 “In Grandfather's Day”, page 3	26.1 KB
`page-15-4.mp3`	Story page	Level 15 “In Grandfather's Day”, page 4	27.0 KB
`page-15-5.mp3`	Story page	Level 15 “In Grandfather's Day”, page 5	30.5 KB
`page-16-1.mp3`	Story page	Level 16 “You Can't Always Tell”, page 1	35.3 KB
`page-16-2.mp3`	Story page	Level 16 “You Can't Always Tell”, page 2	31.8 KB
`page-16-3.mp3`	Story page	Level 16 “You Can't Always Tell”, page 3	29.7 KB
`page-16-4.mp3`	Story page	Level 16 “You Can't Always Tell”, page 4	30.7 KB
`page-16-5.mp3`	Story page	Level 16 “You Can't Always Tell”, page 5	37.2 KB
`page-17-1.mp3`	Story page	Level 17 “Shuffle to Buffalo”, page 1	26.1 KB
`page-17-2.mp3`	Story page	Level 17 “Shuffle to Buffalo”, page 2	20.8 KB
`page-17-3.mp3`	Story page	Level 17 “Shuffle to Buffalo”, page 3	24.6 KB
`page-17-4.mp3`	Story page	Level 17 “Shuffle to Buffalo”, page 4	26.8 KB
`page-17-5.mp3`	Story page	Level 17 “Shuffle to Buffalo”, page 5	25.3 KB

`page-18-1.mp3`	Story page	Level 18 “In Grandfather's Day”, page 1	27.0 KB
`page-18-2.mp3`	Story page	Level 18 “In Grandfather's Day”, page 2	27.1 KB
`page-18-3.mp3`	Story page	Level 18 “In Grandfather's Day”, page 3	32.3 KB
`page-18-4.mp3`	Story page	Level 18 “In Grandfather's Day”, page 4	22.1 KB
`page-18-5.mp3`	Story page	Level 18 “In Grandfather's Day”, page 5	32.9 KB
`page-19-1.mp3`	Story page	Level 19 “The Boy With the Ball”, page 1	28.9 KB
`page-19-2.mp3`	Story page	Level 19 “The Boy With the Ball”, page 2	15.6 KB
`page-19-3.mp3`	Story page	Level 19 “The Boy With the Ball”, page 3	23.7 KB
`page-19-4.mp3`	Story page	Level 19 “The Boy With the Ball”, page 4	23.5 KB
`page-19-5.mp3`	Story page	Level 19 “The Boy With the Ball”, page 5	21.8 KB
`page-20-1.mp3`	Story page	Level 20 “Sploosh!”, page 1	27.9 KB
`page-20-2.mp3`	Story page	Level 20 “Sploosh!”, page 2	21.6 KB
`page-20-3.mp3`	Story page	Level 20 “Sploosh!”, page 3	27.5 KB
`page-20-4.mp3`	Story page	Level 20 “Sploosh!”, page 4	26.1 KB
`page-20-5.mp3`	Story page	Level 20 “Sploosh!”, page 5	34.9 KB
`page-21-1.mp3`	Story page	Level 21 “First Day at the Factory”, page 1	26.7 KB
`page-21-2.mp3`	Story page	Level 21 “First Day at the Factory”, page 2	22.7 KB
`page-21-3.mp3`	Story page	Level 21 “First Day at the Factory”, page 3	25.2 KB
`page-21-4.mp3`	Story page	Level 21 “First Day at the Factory”, page 4	25.5 KB
`page-21-5.mp3`	Story page	Level 21 “First Day at the Factory”, page 5	29.0 KB
`page-22-1.mp3`	Story page	Level 22 “An Ice Idea”, page 1	30.0 KB
`page-22-2.mp3`	Story page	Level 22 “An Ice Idea”, page 2	23.4 KB

`page-22-3.mp3`	Story page	Level 22 “An Ice Idea”, page 3	19.6 KB
`page-22-4.mp3`	Story page	Level 22 “An Ice Idea”, page 4	29.5 KB
`page-22-5.mp3`	Story page	Level 22 “An Ice Idea”, page 5	32.0 KB
`page-23-1.mp3`	Story page	Level 23 “How to Skateboard”, page 1	25.0 KB
`page-23-2.mp3`	Story page	Level 23 “How to Skateboard”, page 2	23.8 KB
`page-23-3.mp3`	Story page	Level 23 “How to Skateboard”, page 3	22.5 KB
`page-23-4.mp3`	Story page	Level 23 “How to Skateboard”, page 4	26.2 KB
`page-23-5.mp3`	Story page	Level 23 “How to Skateboard”, page 5	27.1 KB
`page-24-1.mp3`	Story page	Level 24 “Anansi and the Cook Pots”, page 1	28.8 KB
`page-24-2.mp3`	Story page	Level 24 “Anansi and the Cook Pots”, page 2	20.3 KB
`page-24-3.mp3`	Story page	Level 24 “Anansi and the Cook Pots”, page 3	31.4 KB
`page-24-4.mp3`	Story page	Level 24 “Anansi and the Cook Pots”, page 4	25.1 KB
`page-24-5.mp3`	Story page	Level 24 “Anansi and the Cook Pots”, page 5	29.1 KB
`page-25-1.mp3`	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 1	32.2 KB
`page-25-2.mp3`	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 2	21.4 KB
`page-25-3.mp3`	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 3	20.0 KB
`page-25-4.mp3`	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 4	25.8 KB
`page-25-5.mp3`	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 5	23.7 KB
`page-26-1.mp3`	Story page	Level 26 “First Fast”, page 1	24.1 KB
`page-26-2.mp3`	Story page	Level 26 “First Fast”, page 2	21.4 KB
`page-26-3.mp3`	Story page	Level 26 “First Fast”, page 3	25.6 KB
`page-26-4.mp3`	Story page	Level 26 “First Fast”, page 4	27.2 KB

`page-26-5.mp3`	Story page	Level 26 “First Fast”, page 5	34.7 KB
`page-27-1.mp3`	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 1	33.9 KB
`page-27-2.mp3`	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 2	21.3 KB
`page-27-3.mp3`	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 3	23.0 KB
`page-27-4.mp3`	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 4	28.6 KB
`page-27-5.mp3`	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 5	28.3 KB
`page-28-1.mp3`	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 1	31.1 KB
`page-28-2.mp3`	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 2	23.9 KB
`page-28-3.mp3`	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 3	36.5 KB
`page-28-4.mp3`	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 4	33.9 KB
`page-28-5.mp3`	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 5	32.6 KB
`page-29-1.mp3`	Story page	Level 29 “Clara Barton: Civil War Hero”, page 1	31.1 KB
`page-29-2.mp3`	Story page	Level 29 “Clara Barton: Civil War Hero”, page 2	24.8 KB
`page-29-3.mp3`	Story page	Level 29 “Clara Barton: Civil War Hero”, page 3	24.6 KB
`page-29-4.mp3`	Story page	Level 29 “Clara Barton: Civil War Hero”, page 4	28.4 KB
`page-29-5.mp3`	Story page	Level 29 “Clara Barton: Civil War Hero”, page 5	36.6 KB
`page-30-1.mp3`	Story page	Level 30 “We Also Serve”, page 1	33.9 KB
`page-30-2.mp3`	Story page	Level 30 “We Also Serve”, page 2	25.2 KB
`page-30-3.mp3`	Story page	Level 30 “We Also Serve”, page 3	34.9 KB
`page-30-4.mp3`	Story page	Level 30 “We Also Serve”, page 4	32.0 KB
`page-30-5.mp3`	Story page	Level 30 “We Also Serve”, page 5	31.9 KB

7 Fonts

Fonts

File	Size	Used in the game	Source	Author	License
`fonts/LuckiestGuy-Regular.woff2`	26 KB	Logo, big titles, scores	Google Fonts: Luckiest Guy	Astigmatic	Apache License 2.0
`fonts/Fredoka-Variable.woff2`	28 KB	Buttons and interface text	Google Fonts: Fredoka	Milena Brandão, Hafontia	SIL Open Font License 1.1
`fonts/Andika-Regular.woff2`	19 KB	Story text, questions, answers	Google Fonts: Andika	SIL International	SIL Open Font License 1.1
`fonts/Andika-Bold.woff2`	19 KB	Bold story text	Same	SIL International	SIL Open Font License 1.1
`fonts/LilitaOne-Regular.woff2`	12 KB	Back-up interface font	Google Fonts: Lilita One	Juan Montoreano	SIL Open Font License 1.1
`fonts/AtkinsonHyperlegible-Regular.woff2`	23 KB	Back-up reading font	Google Fonts: Atkinson Hyperlegible	Braille Institute of America	SIL Open Font License 1.1
`fonts/AtkinsonHyperlegible-Bold.woff2`	23 KB	Back-up reading font, bold	Same	Braille Institute of America	SIL Open Font License 1.1

All fonts are bundled in the app; nothing is loaded from Google at run time.

8 Images

Images

File	Size	Used in the game	Source	Author	License
`ui/mira-hello.png`	29 KB	Teacher Mira waving on the title screen	unknown	unknown	unknown
`ui/mira-talk.png`	16 KB	Mira talking (default mood in the dock)	unknown	unknown	unknown
`ui/mira-happy.png`	20 KB	Mira after a right answer	unknown	unknown	unknown
`ui/mira-think.png`	17 KB	Mira after a wrong answer	unknown	unknown	unknown
`ui/mira-wave.png`	17 KB	Mira greeting	unknown	unknown	unknown
`ui/mira-cheer.png`	14 KB	Mira on the results screen	unknown	unknown	unknown
`ui/mira.png`	15 KB	Mira's small face in the race speech bubble	unknown	unknown	unknown
`ui/mira-full.png`	23 KB	**Not used** by the current code	unknown	unknown	unknown
`ui/backdrop.jpg`	164 KB	Title screen and general background	unknown	unknown	unknown
`ui/world-0.jpg`	111 KB	Sunny Meadow page art and backdrop	unknown	unknown	unknown

`ui/world-1.jpg`	114 KB	River Bend	unknown	unknown	unknown
`ui/world-2.jpg`	137 KB	Story Town	unknown	unknown	unknown
`ui/world-3.jpg`	113 KB	Palm Beach	unknown	unknown	unknown
`ui/world-4.jpg`	106 KB	Summit Peak	unknown	unknown	unknown
`ui/star.png`	7 KB	Earned star	unknown	unknown	unknown
`ui/star-off.png`	7 KB	Empty star	unknown	unknown	unknown
`icons/icon-192.png`	16 KB	Browser tab icon	unknown	unknown	unknown
`icons/icon-512.png`	37 KB	Large app icon	unknown	unknown	unknown
`res/mipmap-*/ic_launcher*.png`	—	Android launcher icon (all sizes)	unknown	unknown	unknown
Mira's small SVG face, map, coins, buttons, sky, scenery made in code	—	Drawn in HTML, CSS, SVG and three.js code	SummaRace code	SummaRace developer(s)	Part of the game

9 Code libraries and data

Code and data

File	Size	Used in the game	Source	Author	License
`three.min.js`	589 KB	3D engine for the race	github.com/mrdoob/three.js (release r128)	three.js authors	MIT
`GLTFLoader.js`	94 KB	Loads the GLB models	three.js examples (`examples/js/loaders/GLTFLoader.js`), exact release not recorded	three.js authors	MIT
`words.js`	195 KB	24,887 common English words for the summary spelling check	wordfreq word list	Robyn Speer	CC BY-SA 4.0 (data); the file is shared under the same license
`stories.js`	174 KB	The 30 stories, questions and story parts	Research team's study materials. Wrong choices, race wording and model summaries were written for the game	Research team (client); game additions by the developer(s)	Not stated. Original story authors are not listed in the file
`voice/voice.js`	18 KB	Index from Mira's lines to voice clips	Generated by `build-voice.sh` (lost)	SummaRace developer(s)	Part of the game
`index.html`	288 KB	The whole game: screens, rules, race	Written for SummaRace	SummaRace developer(s)	Not stated (no LICENSE file in the repo)

The Android app (`MainActivity.java`) uses only the Android SDK. There are no third-party Android libraries.

10 Tools used to make assets

Tool	Used for	Link
glTF-Transform 4.1.1	Compressing the GLB models (stamped in each file)	gltf-transform.dev
UnityGLTF	Exporter stamped inside the coin and cloud files	github.com/KhronosGroup/UnityGLTF
Kokoro TTS	Mira's voice clips	huggingface.co/hexgrad/Kokoro-82M
FFmpeg (Lavf 60.16)	Stamped as the encoder in the music files	ffmpeg.org

PART 8 · HANDOVER

Handover and Known Open Items

State of the project on 2026-11-06, when the game was declared finished.

1 What is handed over

Item	Where
Source code (game + Android app)	github.com/orbitsdev/summaracev3, branch `main`, version 1.3.6 (versionCode 17)
These documents	`docs/` in the repo (separate pages) and this compiled file (HTML and PDF)
Release signing key	Kept on the owner's main PC only, never in the repo. Every future update must be signed with this same key.
Next version number	versionCode 18 or higher

2 Known open items

Item	Why it matters	Status
1.3.6 release build not yet tested on a phone	The lane-answer race (PR #6) and race UI changes (PR #7) were merged before a full phone test. A signed release build should be played through on a real phone before it goes to the client.	To do
PR #4: race music speeds up	Optional. Merge or close.	Owner's call
PR #5: Story Stop voice lines	No longer needed since the race uses lane answers. Close it.	Close
Music and UI images have no recorded source	Needed for honest credits. See Part 7, section 2.	Owner to confirm
wordfreq (CC BY-SA 4.0) not credited in the game	Its license requires a credit line on the credits screen.	To do
Credits screen points to a lost CREDITS.md	Point it at these documents or ship a new credits file.	To do
Font license texts not shipped	OFL and Apache 2.0 ask for the license text to go with the fonts.	To do
Story authors not credited	Some stories are published books. The research team should confirm permission and credit lines.	Client to confirm
Two stories titled "In Grandfather's Day" (levels 15 and 18)	Confusing on the map and in the CSV.	Client to confirm
10+ Mira lines have no voice clip	They use the phone's robot voice. The clip build script was lost.	Known limit
Same name + section on one phone share a profile	Two learners with the same name in one section would mix their progress.	Known limit

SummaRace v3 · compiled from `docs/` in the repo · describes `main` at version 1.3.6.